Stony Brook Institute
at Anhui University

安徽大学纽约石溪学院

Chapter 2 Random variable and their probability distributions

Section 2.6 Conditional distribution

- Discrete conditional distribution
- Continuous conditional distribution

Section 2.6 Conditional distribution

In Chapter 1, the concept of conditional probability has been introduced, that is, the conditional probability of event ***A*** under the condition of event ***B***:

$$P(A \mid B) = \frac{P(AB)}{P(B)}$$

Now there are two random variables ***X*** and ***Y***. if Q: under the condition that ***Y*** takes some or some values, find the probability distribution of random variable ***X***.

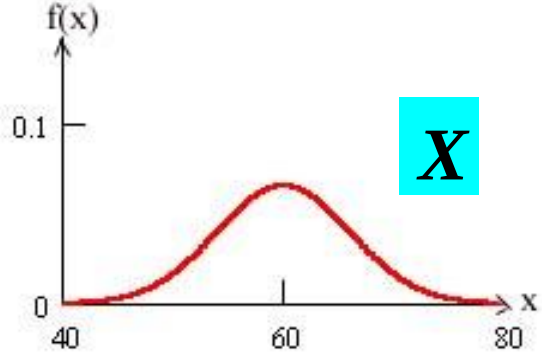
Conditional distribution

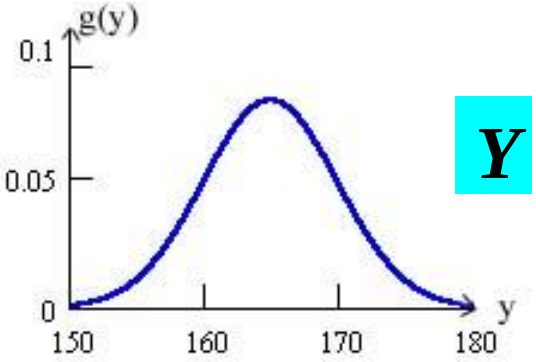
Section 2.6 Conditional distribution

Question: Suppose X and Y are weight and height of a person among these people. X and Y are random variables has its own distributions. So what is the distribution of X when Y is limited to 1.7 ~ 1.8 meters.

Weight *X*

Height *Y*





Discrete conditional distribution

Definition 2.6.1: Suppose (X,Y) is a discrete two-dimensional $r.v.$, and its joint probability distribution is $P\{X = xi, Y = y_j\} = p_{ij}$. We know that

$$p_{i\cdot} = \sum_{j=1}^{\infty} p_{ij} = P\{X = x_i\}, i = 1,2, \ldots$$

and

$$p_{\cdot j} = \sum_{i=1}^{\infty} p_{ij} = P\{Y = y_j\}, j = 1,2, \ldots$$

If $P\{Y = y_j\} = p_{\cdot j} > 0$, we call

$$P\{X = xi | Y = y_j\} = \frac{P\{X = xi, Y = y_j\}}{P\{Y = y_j\}} = \frac{p_{ij}}{p_{\cdot j}}$$

conditional probability distribution of *X* when Y = y_j is given.

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$$P\{Y = y_j | X = xi\} = \frac{P\{X = xi, Y = y_j\}}{P\{X = xi\}} = \frac{p_{ij}}{p_{i\cdot}}$$

conditional probability distribution of Y when $X = xi$ is given.

Section 2.6 Conditional distribution

Properties of conditional distribution

$$1^0. \quad P(X = x_i \mid Y = y_j) \geq 0$$

$$2^0. \quad \sum_{i=1}^{\infty} P(X = x_i \mid Y = y_j) = \sum_{i=1}^{\infty} \frac{P_{ij}}{P_{.j}}$$

$$= \frac{1}{P_{.j}} \sum_{i=1}^{\infty} P_{ij} = \frac{1}{P_{.j}} \cdot P_{.j} = 1$$

$$3^0. \quad \sum_{j=1}^{\infty} P(Y = y_j \mid X = x_i) = 1$$

Discrete conditional distribution

Example 2.6.1: In a motor factory, two key processes are done by machine. One is tightening three screws, another one is welding two spots. Let X be the number of screws that are not tightened, Y be the number of bad welding spot. According to the accumulated data we know the distribution of (X,Y) is:

$Y \backslash X$	0	1	2	3	$P\{Y=j\}$
0	0.840	0.030	0.020	0.010	0.900
1	0.060	0.010	0.008	0.002	0.080
2	0.010	0.005	0.004	0.001	0.020
$P\{X=i\}$	0.910	0.045	0.032	0.013	1.000

(1) Find the conditional distribution law of Y when X=1
(2) Find the conditional distribution law of X when Y=0

Discrete conditional distribution

Solution: (1)

$$P\{Y=0|X=1\}=\frac{P\{X=1,Y=0\}}{P\{X=1\}}=\frac{0.030}{0.045},$$

$$P\{Y=1|X=1\}=\frac{P\{X=1,Y=1\}}{P\{X=1\}}=\frac{0.010}{0.045},$$

$$P\{Y=2|X=1\}=\frac{P\{X=1,Y=2\}}{P\{X=1\}}=\frac{0.005}{0.045},$$

the conditional distribution law of Y when X=1 is

$Y=k$	0	1	2
$P\{Y=k\|X=1\}$	$\frac{6}{9}$	$\frac{2}{9}$	$\frac{1}{9}$

(2) the conditional distribution law of X when Y=0 is

$X=k$	0	1	2	3
$P\{X=k\|Y=0\}$	$\frac{84}{90}$	$\frac{3}{90}$	$\frac{2}{90}$	$\frac{1}{90}$

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Section 2.6 Conditional distribution

- In the discrete r.v., the conditional distribution is derived from the conditional probability, thus $P(X = x) > 0,\ P(Y = y) > 0$.
- However, it is noted that in the discussion of one-dimensional random variables, it is pointed out that one of the fundamental differences between continuous and discrete random variables is that for continuous random variables, $P(X=x_i)=0$, $P(Y=y_i)= 0$.
- Therefore, the concept of conditional distribution cannot be derived from conditional probability for continuous random variables.
- So, the concept of conditional distribution must be derived from the method of distribution function and limit.

Continuous conditional distribution

$$F_{X|Y}(x|y) = P(X \le x | Y = y)$$

$$= \lim_{\varepsilon \to 0^+} P(X \le x | y - \varepsilon < Y \le y + \varepsilon)$$

$$= \lim_{\varepsilon \to 0^+} \frac{P(X \le x, y - \varepsilon < Y \le y + \varepsilon)}{P(y - \varepsilon < Y \le y + \varepsilon)}$$

$$= \lim_{\varepsilon \to 0^+} \frac{F(x, y + \varepsilon) - F(x, y - \varepsilon)}{F_Y(y + \varepsilon) - F_Y(y - \varepsilon)}$$

$$= \lim_{\varepsilon \to 0^+} \frac{[F(x, y + \varepsilon) - F(x, y - \varepsilon)]/2\varepsilon}{[F_Y(y + \varepsilon) - F_Y(y - \varepsilon)]/2\varepsilon}$$

Continuous conditional distribution

$$F_{X|Y}(x|y) = P(X \le x | Y = y)$$

$$= \lim_{\varepsilon \to 0^+} \frac{[F(x, y+\varepsilon) - F(x, y-\varepsilon)]/2\varepsilon}{[F_Y(y+\varepsilon) - F_Y(y-\varepsilon)]/2\varepsilon}$$

$$= \frac{\dfrac{\partial F(x,y)}{\partial y}}{\dfrac{dF_Y(y)}{dy}} = \frac{\dfrac{\partial}{\partial y}\left[\int_{-\infty}^{x}\int_{-\infty}^{y} f(x,y)dxdy\right]}{f_Y(y)}$$

$$= \frac{\int_{-\infty}^{x} f(x,y)dx}{f_Y(y)} = \int_{-\infty}^{x} \frac{f(x,y)}{f_Y(y)}dx$$

Continuous conditional distribution

$$F_{X|Y}(x|y) = P(X \le x|Y = y) = \int_{-\infty}^{x} \frac{f(x,y)}{f_Y(y)} dx$$

According to definition：

$$F_{X|Y}(x|y) = \int_{-\infty}^{x} f_{X|Y}(x|y)\, dx$$

So , $$f_{X|Y}(x|y) = \frac{f(x,y)}{f_Y(y)}$$

Similarly, $$f_{Y|X}(y|x) = \frac{f(x,y)}{f_X(x)}$$

Continuous conditional distribution

Definition Let (X, Y) be a continuous two-dimensional r.v., and its joint pdf is $f(x, y)$. For any fixed y, and $f_Y(y) \neq 0$, we call $\frac{f(x,y)}{f_Y(y)}$ the conditional pdf of X when $Y = y$ is given, and denote

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)}$$

We call

$$\int_{-\infty}^{X} f_{X|Y}(x|y)\mathrm{d}x = \int_{-\infty}^{x} \frac{f(x, y)}{f_Y(y)}\mathrm{d}x$$

the conditional *cdf* of X when $Y = y$ is given, and denote

$$F_{X|Y}(x|y) = P\{X \leq x|Y = y\} = \int_{-\infty}^{x} \frac{f(x, y)}{f_Y(y)}\mathrm{d}x$$

Continuous conditional distribution

Similarly, For any fixed x, and $f_X(x) > 0$, we call $\frac{f(x,y)}{f_X(x)}$ the conditional pdf of Y when $X = x$ is given, and denote

$$f_Y(y|x) = \frac{f(x,y)}{f_X(x)}, f_X(x) \neq 0$$

We call $\int_{-\infty}^{y} f_{\mathrm{Y|X}}(y|x)\mathrm{d}y = \int_{-\infty}^{y} \frac{f(x,y)}{f_X(x)}$ the conditional pdf of Y when $X = x$ is given, and denote

$$F_{Y|X}(y|x) = P\{Y \leq y|X = x\} = \int_{-\infty}^{y} \frac{f(x,y)}{f_X(x)}\mathrm{d}y$$

Continuous conditional distribution

Example 2.6.2: G is a circle in the xOy with an radius of A, and center of G is O, if the PDF of r.v. (X,Y) is

$$f(x,y)=\begin{cases}\frac{1}{\pi}, & (x,y)\in G\\ 0, & others\end{cases}$$

Find the conditional PDF $f_{X|Y}(x|y)$.

Solution:

$$f_Y(y)=\int_{-\infty}^{\infty} f(x,y)\,\mathrm{d}\,x$$

$$=\begin{cases}\boldsymbol{\frac{1}{\pi}\int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}}\mathrm{d}\,x=\frac{2}{\pi}\sqrt{1-y^2},\ -1\le y\le 1,}\\ \boldsymbol{0,\qquad others.}\end{cases}$$

$\boldsymbol{when\ -1<y<1\ ,we\ have}$

$$\boldsymbol{f_{X|Y}(x|y)=\begin{cases}\frac{1/\pi}{(2/\pi)\sqrt{1-y^2}}=\frac{1}{2\sqrt{1-y^2}}, & -\sqrt{1-y^2}\le x\le\sqrt{1-y^2},\\ 0, & others.\end{cases}}$$

Continuous conditional distribution

Example 2.6.3: X is a uniform r.v. between 0 and 1.When X=x, (0<x<1), Y is a uniform r.v. between x and 1, find the PDF of Y, i.e. $f_Y(y)$

Solution: The PDF of X is

$$f_X(x) = \begin{cases} 1, & 0 < x < 1, \\ 0, & others. \end{cases}$$

When X=x, then conditional PDF of Y is

$$f_{Y|X}(y|x) = \begin{cases} \dfrac{1}{1-x}, & 0 < x < y < 1, \\ 0, & others. \end{cases}$$

The joint PDF of r.v. X and Y is

$$f(x,y) = f_{Y|X}(y|x) f_X(x) = \begin{cases} \dfrac{1}{1-x}, & 0 < x < y < 1, \\ 0, & others. \end{cases}$$

The marginal PDF of Y respect to (X,Y) is

$$f_Y(y) = \int_{-\infty}^{\infty} f(x,y)\,\mathrm{d}\,x = \begin{cases} \displaystyle\int_0^y \frac{1}{1-x}\,\mathrm{d}\,x = -\ln(1-y), 0 < y < 1, \\ 0, \qquad\qquad\qquad\qquad others. \end{cases}$$

Continuous conditional distribution

Examples

Example 2.6.4 : Suppose that the pdf of (X,Y) is :

$$f(x,y)$$

$$= \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)}\left[\frac{(x-\mu_1)^2}{\sigma_1^2} - 2\rho\frac{(x-\mu_1)(x-\mu_2)}{\sigma_1\sigma_2} + \frac{(y-\mu_2)^2}{\sigma_2^2}\right]}$$

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma_1} e^{-\frac{(x-\mu_1)^2}{2\sigma_1^2}}, \quad f_Y(y) = \frac{1}{\sqrt{2\pi}\sigma_2} e^{-\frac{(x-\mu_2)^2}{2\sigma_2^2}}$$

Find : $f_{X|Y}(x|y)$

Continuous conditional distribution

Examples

Solution:

$$f_{X|Y}(x|y) = \frac{f(x,y)}{f_Y(y)}$$

$$= \frac{1}{\sqrt{2\pi}\sigma_1\sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)}[\frac{x-\mu_1}{\sigma_1}-\rho\frac{y-\mu_2}{\sigma_2}]^2}$$

Obviously, it also obeys the normal distribution:

$$N\left(\mu_1 + \rho\frac{\sigma_1}{\sigma_2}(y-\mu_2),\ (1-\rho^2)\sigma_1^2\right)$$

The marginal distribution and conditional distribution of normal distribution still obey normal distribution.